NSW INDEPENDENT TRIAL EXAMS – 2020
MATHEMATICS STANDARD YEAR 11 EXAM
MARKING GUIDELINES

Section I

Question	Answer	Assessed Outcome	Band
1.	D	MS-M1.1, MS11-3	3
2.	B	MS-S2, MS11-9	3
3.	C	MS-M1.1, MS11-3	3
4.	D	MS-S1, MS11-9	3
5.	D	MS-F1.2, MS11-5	3
6.	C	MS-A1, MS11-1	3
7.	B	MS-M1.3, MS11-4	3
8.	C	MS-A1, MS11-1	3
9.	A	MS-A2, MS11-2	4
10.	D	MS-M1.1, MS11-9	4
11.	C	MS-M2, MS11-3	4
12.	C	MS-S1, MS11-9	4
13.	B	MS-M1.1, MS11-4	4
14.	C	MS-M1.2, MS11-4	4
15.	C	MS-F1.3, MS11-9	4
16.	B	MS-S1.1, MS11-9	4
17.	A	MS-S1.2, MS11-7	5
18.	D	MS-S2, MS11-10	5
19.	A	MS-F1.2, MS11-6	5
20.	A	MS-M1.2, MS11-3	6

Section II

Part	Answer	Mark	Outcome Assessed	Band
21	**Categorical, Ordinal**	1	MS-S1.1, MS11-10	3
22	Standard deviation = **13.67**	1	MS-S1.2, MS11-9	3
23	The difference in time between the two cities is 19 hrs **Sydney is ahead of San Francisco by 19 hours.** Time in Sydney is 1:15 pm Sun 5^{th} + 19 hours **i.e. 8:15 am Monday 6^{th}.**	 1 1	MS-M2, MS11-9	3
24	**1.50 kg = 1500 g** Number of grains of rice $= 1500 \div 0.00171$ $= 877\ 000$ $= \mathbf{8.8 \times 10^5}$	1 1	MS-S1.3, MS11-3	3
25	From the formula $m = h^2 \times b$ When $b = 21.5$, $m = 1.7^2 \times 21.5$ $= \mathbf{62.135}$ When $b = 24.5$, $m = 1.7^2 \times 24.5$ $= \mathbf{70.805}$ **Average weight** $= \mathbf{(62.135 + 70.805) \div 2}$ $= \mathbf{66.47}$ **kg**	 1 1	MS-A1, MS11-6	3
26	\$62 500 = \$45 000 + \$17 500 Stamp duty = **(0.03 × \$45 000) + (0.05 × \$17 500)** **= \$2225**	 1 1	MSF1.3, MS11-10	3
27	One possible set: {5, 8, 8, 12, 13} **(Note: 1 mark for largest score of 13** **1 mark for mode of 8 with median 8)** Another possible set: {5, 6, 8, 8, 13)	2	MS-S1.2, MS11-10	4
28(a)	$BAC = \dfrac{10(3) - 7.5(2)}{5.5(42)}$ $= \dfrac{30 - 15}{231} = 0.06$	1 1	MS-A1, MS11-9	3
28(b)	$D = \dfrac{0.06}{0.015}$ $= 4$ hours BAC is expected to reach zero at 10:00 pm + 4 hours **i.e. at 2:00 am the next morning**	 1	MS-A1, MS11-10	3
29	$\dfrac{5(F - 32)}{9} = C$ $\mathbf{5(F - 32) = 9C}$ $F - 32 = \dfrac{9C}{5}$ $\mathbf{F = \dfrac{9C}{5} + 32}$	 1 1 1	MS-A1, MS11-1	4
30	$\mathbf{7 - 3p = 4p}$ $7p = 7$ $\mathbf{p = 1}$	1 1	MS-A1, MS11-1	4

31	Maximum allowance/fortnight $= \$455.20$ Tony's parents earn $78 100/annum Allowance reduces by ($78 100 – $53 728) × 0.20 = $24372 × 0.20/annum = $4874.40/annum **= $187.48/fortnight** So, Tony receives $455.20 – $187.48 **= $267.72/fortnight**	1 1 1	MS-F1.2, MS11-9	4
32	**Surface area** $= 2\pi r^2 + 2\pi rh$ $= (2 \times \pi \times 10^2) + (2 \times \pi \times 10 \times 20)$ $= 200\pi + 400\pi$ $= 600\pi$ $= 1885$ **cm²**	1 1	MS-M1.2, MS11-3	3
33	Required area = area of large semicircle of radius 9 cm – area of one small semicircle of radius 3 cm $= (\frac{1}{2} \times \pi \times 9^2) - (\frac{1}{2} \times \pi \times 3^2)$ $= \frac{81\pi}{2} - \frac{9\pi}{2}$ $= 36\pi$ **= 113 cm²**	 1 1	MS-M1.2, MS11-4	4
34	After 2 days, the pool has 65 000 × 0.975 × 0.975 L **= 61 790.625 L** Additional water added is 65 000 – 61 790.625 L **= 3209.375 L** Percentage of water added $= \frac{3209.375 \times 100}{61\,790.625}$ **= 5.2%**	 1 1 1	MS-F1.1, MS11-6	5
35	**Interest earned per year** $= \frac{\$680}{4} = \$170$ **$170 is 5% of amount invested** **Amount invested = $170 × 20** **= $3400**	1 1	MS-F1.1, MS11-9	5
36(a)	The median **increased** (from 75 beats/min to 90 beats/min)	1	MS-S1.2-MS11-7	3
36(b)	The I.Q.R **reduced** (from approx. 15 to 8)	1	MS-S1.2, MS11-7	3
36(c)	**25% of 40 = 10 people**	1	MS-S1.2, MS11-7	3
36(d)	An outlier would be a data point $> Q_3 + (1.5 \times IQR)$ **In this case, a data point > 93 + (1.5 × 8)** **= 105** **So, 115 beats per minute** would be considered an outlier for the data presented after exercise	1 1	MS1.2, MS11-10	4
37(a)	The gradient of the line is 9/8 = 1.125 and the y-intercept is zero. The equation is $y = mx + b$ i.e. $y = 1.125x$	1 1	MS-A2, MS11-10	4
37(b)	The gradient of 1.125 indicates the rate which is equivalent to **a tax of 12.5% added to prices.**	1	MS-A2, MS11-6	4
37(c)	The point (c) indicates x = 6, so using y = 1.125x The y-coordinates is 6 × 1.125 = $6.75 **Coordinates of (c) are (6, 6.75)**	 1	MS-A2, MS11-6	4
37(d)	**112.5% = $150** **12.5% = ($150 ÷ 112.5) × 12.5** **= $16.67** **A tax of $16.67 was added to the original price.**	1 1	MS-F1.1, MS11-6	5

38(a)	Taxable income = \$112 525.00 – **((\$135 × 26) + \$220 + (0.125 × \$2850) + (\$15 × 12) + 2820 × \$0.37)** = \$112 525.00 – (\$3510 + \$220 + \$356.25) + \$180 + \$1043.40)) = \$112 525.00 – (\$5309.65) = **\$107 215** (to the nearest dollar)	**1** **1**	MS-F1.2, MS11-9	4
38(b)	Medicare Levy = 0.02 × \$107 215 **= \$2144.30**	**1**	MS-F1.2, MS11-9	4
38(c)	**Tax payable on \$107 215:** = \$19 822 + (0.37 × (\$107 215 – \$87 000) = \$19 822 + (\$7479.55) **= \$27 301.55** **Total tax payable:** **= \$27 301.55 + \$2144.30** **= \$29 445.85** Tax paid (PAYG) = \$30 250 **So, Brendon is entitled to a refund of:** **\$30 250 – \$29 445.85** **= \$804.15**	 **1** **1** **1**	MS-F1.2, MS11-9	5
39	$\text{Let } 10 = \sqrt{(x_2 - -3)^2 + (-2-4)^2}$ $= \sqrt{(x_2+3)^2 + (-6)^2}$ $= \sqrt{(x_2+3)^2 + 36}$ $100 = (x_2+3)^2 + 36$ $(x_2+3)^2 = 64$ $x_2 + 3 = +/-8$ $x_2 = 5 \text{ or } -11$	 **1** **1** **1**	MS-A1, MS11-10	5
40(a)	Area PQRS = ½ × (6 + 2) × (10 + 6) **= 64 cm²**	**1**	MS-M1.2, MS11-4	3
40(b)	Area ΔQRT = Area $PQRS$ – (area ΔPQT + area ΔSTR) = 64 – ((½ × 2 × 6) + (½ × 6 × 10)) = 64 – (36) **= 28 cm²**	**1** **1**	MS-M1.2, MS11-4	4
40(c)	Required percent is 28/64 × 100% **= 43.75%**	**1** **1**	MS-M1.1, MS11-3	3
41(a)	$\tfrac{3}{4} \text{ of } 2\pi r = 12\pi$ $\dfrac{3r}{2} = 12$ $r = (24 \div 3)$ $= 8 \text{ cm}$	**1** **1**	MS-M1.2, MS11-9	4
41(b)	**Using Pythagoras' Theorem with r = 8** $PQ^2 = 8^2 + 8^2$ $= 128$ $PQ = 11.31 \text{ cm}$	 **1** **1**	MS-M1.2, MS11-9	3

42(a)	**$0.4 \times 0.4 = 0.16 = 16\%$**	1	MS-S2, MS11-10	3
42(b)	**$(0.4 \times 0.6) \times (0.6 \times 0.4)$** **$= 0.24 + 0.24$** **$= 0.48$** **$= 48\%$**	1 1	MS-S2, MS11-10	3
42(c)	**1 – Pr (both monkeys are of species A)** $= 1 - (0.16)$ using result of (a) **$= 0.84$** **$= 84\%$**	1 1	MS-S2, MS11-10	4
43(a)	The highest rate of depreciation is represented by the line with the greatest slope. In this case, **Machine 1 has the highest rate.**	1	MS-A2, MS11-6	3
43(b)	Machine 2 depreciates by $\$3500 \div 7$/year i.e. \$500/year **Machine 2 has 2 years remaining when Machine 1 has no value, so its remaining value will be \$1000.**	1	MS-A2, MS11-6	5
43(c)	Equations of depreciation are: M_1: $V_1 = 4000 - 800n$ M_2: $V_2 = 3500 - 500n$ When $V_1 = V_2$, $4000 - 800n = 3500 - 500n$ $300n = 500$ $n =$ **5/3 years** **= 20 months (1yr, 8 months)**	1 1 1	MS-A2, MS11-9	5
44	Fuel consumption is 8.5 L/100 km The tank has the equivalent of 90 km remaining i.e. $\frac{8.5}{100} \times 90 =$ **7.65 litres** The tank requires 60 – 7.65 = **52.35** litres to fill to capacity Cost of fuel = $52.35 \times \$1.72$ **= \$90**	 1 1 1	MS-F1.3, MS11-10	5/6
45(a)	Sum of edges is $4y + 4x + 8x$ **$= 4y + 12x$** Let $4y + 12x = 36$ **$4y = 36 - 12x$** $y = 9 - 3x$	 1 1	MS-M1.2, MS11-3	5/6
45(b)	Volume $= 2x^2y$ $= 2x^2(9 - 3x)$ using (a) $= 18x^2 - 6x^3$	 1 1	MS-M1.2, MS11-1, MS-A1, MS11-3	5
45(c)	From (b), $V = 6x^2(3 - x)$ if $x = 0$, Volume (V) = 0 and if $x = 3$, $V = 0$ Any value of $x > 3$ will give negative V and x must be > 0 So x must lie between 0 and 3 Thus, $y = 9 - 3x$ must lie between 0 (if $x = 3$) and 9 (if $x =0$)	 1 1	MS-M1.3, MS11-10	6